INFRASTRUCTURE RELIABILITY ANALYTICS

Predictive Modelling of Environmental Impacts on UK Railway Operations

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Version
2.0 — Rebuilt Model (2026)

Programme
MSc Data Engineering

Repository
infrastructure-reliability-analytics

Institution
Glasgow Caledonian University

Data Sources
ORR, Met Office, Visual Crossing

Abstract

Rail transport is the backbone of the United Kingdom's economic infrastructure. The network carries approximately 1.7 billion passenger journeys annually and supports billions of pounds of economic activity, yet its reliability remains acutely sensitive to environmental conditions (Office of Rail and Road, 2023). This study presents a data-driven investigation into the relationship between weather variables and train service disruptions, using Scotland's railway network as a representative case study with findings that carry direct implications for rail operations across the whole of the United Kingdom.

The project integrates quarterly train performance data from the Office of Rail and Road (ORR) with historical weather records sourced from the Met Office and Visual Crossing, covering a study period from 2014 to 2026. Six machine learning regression models were trained and evaluated: Linear Regression, Lasso Regression, Ridge Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN). The original 2024 model iteration established baseline performance metrics and identified core data limitations. The 2026 rebuilt model addresses those limitations through improved feature engineering, quarterly aggregation methodology, and more rigorous evaluation procedures.

Key findings confirm statistically significant correlations between adverse weather and train service cancellation scores. K-Nearest Neighbors emerged as the best-performing model with an RMSE of 577.93 and a positive R^2 of 0.1583, with Lasso and Ridge Regression also returning positive R^2 values. Cloud cover and humidity emerged as the strongest environmental predictors from feature importance analysis, alongside precipitation and wind variables. Two models, Random Forest and Decision Tree, returned negative R^2 values, consistent with known overfitting behavior on small datasets. This study proposes a clear pathway for model improvement, with direct applications for the UK Department for Transport, Network Rail, and train operating companies.

Keywords: *railway reliability, machine learning, environmental impact, predictive analytics, UK rail infrastructure, data engineering*

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1. Introduction

The United Kingdom's railway network is one of the most intensively used in Europe. According to the Office of Rail and Road (2023), over 20,000 train services run across the network every weekday, and the system directly underpins the economic productivity of every major UK city. Yet disruption to this network is both frequent and costly. The Department for Transport (2023) estimated that weather-related rail delays and cancellations cost the UK economy hundreds of millions of pounds annually through lost productivity, missed connections, and reactive operational expenditure.

Scotland's railway network, operated principally by ScotRail and managed by Network Rail's Scotland Route, presents a particularly challenging operational environment. Scotland experiences significantly more adverse weather than most of England, including higher annual rainfall, stronger average wind speeds, and more frequent extreme weather events (Met Office, 2023). As climate change intensifies the frequency and severity of such events, the challenge of maintaining reliable rail operations in Scotland and by extension across the UK is only going to grow (Committee on Climate Change, 2021).

This project was originally developed in 2024 as part of an MSc Data Engineering programme at Glasgow Caledonian University. It is now being rebuilt and extended in 2026 with the following aims: to improve the predictive accuracy of the original models, to strengthen the methodological rigor of the analysis, and to draw out the policy implications of the findings for the UK government and rail industry.

1.1 Research Questions

1. Do statistically significant correlations exist between environmental conditions and train service disruptions on Scotland's railway network?
2. Which environmental variables are the strongest predictors of service cancellations?
3. How accurately can machine learning models forecast train service disruptions using environmental data alone, and what additional data would be required to achieve operationally useful accuracy?
4. How do the rebuilt 2026 models compare to the original 2024 baseline in terms of predictive performance?

1.2 Significance of This Research

The significance of this work extends well beyond Scotland. Network Rail is a government-owned company, and its performance directly affects the UK Treasury's investment decisions in transport

infrastructure (HM Treasury, 2021). The Williams-Shapps Plan for Rail (Department for Transport, 2021) set out a vision for Great British Railways, a unified system requiring a level of data-driven operational intelligence that this kind of predictive modelling directly supports.

UK Policy Relevance

The UK government has committed to achieving net-zero carbon emissions by 2050, with rail electrification and a modal shift from road to rail central to that target (Department for Transport, 2021). Unreliable rail services undermine passenger confidence and reduce modal shift. Predictive maintenance and disruption-forecasting tools of the kind developed in this project are therefore not merely academic; they are a prerequisite for delivering the government's transport and climate agenda.

2. Literature Review

2.1 Weather and Railway Reliability

The relationship between weather and railway performance is well-established in the academic literature. Ferranti et al. (2016) demonstrated that heat-related failures on south-east England's railway network were directly traceable to specific temperature thresholds, and proposed that proactive speed restrictions and track inspections could significantly reduce heat-related delays. Their work established a methodological precedent for the kind of threshold-based, weather-driven analysis applied in this project.

Jaroszweski, Chapman and Petts (2010) argued that the transport sector has been slow to account for climate variability in infrastructure planning, and that interdisciplinary approaches combining climate science with operational data are essential for building resilient networks. This view is directly reflected in the multi-source data strategy employed in this project, which combines ORR performance statistics with meteorological data from two independent providers.

Network Rail's own Weather Resilience and Climate Change Adaptation reports (Network Rail, 2022) acknowledge that weather is the single largest external cause of train delays in the UK, responsible for approximately 15% of all recorded delay minutes. The report identifies high winds, flooding, and extreme heat as the three most operationally disruptive weather categories, findings that informed the feature selection strategy in this study.

2.2 Machine Learning in Transport Infrastructure

The application of machine learning to transport reliability prediction has grown significantly over the past decade. Predictive models using regression, ensemble methods, and neural networks have been applied to road networks, aviation, and maritime transport with varying degrees of success (Comi et al., 2020). In the railway domain, Random Forest and Gradient Boosting models have shown particular promise for fault classification when rich operational data is available (Yuan et al., 2019).

However, a consistent finding across the literature is that model performance is tightly coupled to data quality and richness (Pedregosa et al., 2011). Models trained on environmental data alone consistently underperform those that incorporate operational variables such as maintenance history, infrastructure age, and rolling stock condition. This finding directly contextualises the negative R^2 values observed in this study and frames the data limitation challenge as a solvable, rather than fundamental, constraint.

2.3 Gap in the Literature

Despite the breadth of existing work on weather-railway interactions, there is a notable gap in studies that specifically address Scotland's rail network using a combined environmental and operational data approach. Most UK-focused studies concentrate on the busier south-east England network, where data availability is greater. This project partially fills that gap and lays the groundwork for a more comprehensive Scottish rail reliability model as data access improves.

3. Data Sources and Preprocessing

3.1 Data Sources

Two primary datasets were used in this analysis:

Table 1: Data Sources

Dataset	Source	Key Variables
Train Performance Metrics	Office of Rail and Road (ORR)	Trains planned, cancellation score, infra network score, infra external score, operator fault score, total cancellations 2014 quarterly granularity.
Environmental Data	Met Office / Visual Crossing	Max/min/avg temperature, precipitation, humidity, wind gust, wind speed, wind direction, visibility, cloud cover 2014 monthly granularity

The ORR dataset is publicly available through the regulator's official statistical releases and represents the most authoritative source of train performance data in the United Kingdom (Office of Rail and Road, 2023). Environmental data was sourced from two providers to ensure completeness: The Met Office, which is the UK's national meteorological service, and Visual Crossing, a commercial weather data platform providing historical records with high spatial coverage (Met Office, 2023; Visual Crossing Corporation, 2024).

3.2 Data Preprocessing

The preprocessing pipeline involved the following key steps, implemented in Python using the pandas library (McKinney, 2010):

- Date parsing: All date and datetime columns were parsed using pandas.to_datetime() with errors='coerce' to handle any malformed entries gracefully.
- Temporal alignment: The environmental dataset contained monthly observations, while the ORR dataset used quarterly periods. Monthly environmental records were aggregated to quarterly averages using pandas groupby, ensuring both datasets shared the same temporal granularity before merging.
- Dataset merge: The two datasets were joined on a shared quarter key (e.g., 2018Q3), producing a merged dataset of quarterly observations spanning the study period.
- Feature scaling: StandardScaler from scikit-learn (Pedregosa et al., 2011) was applied to all input features prior to model training. This is particularly important for distance-sensitive algorithms such as KNN and regularisation-based models such as Lasso and Ridge Regression.

- Train-test split: An 80/20 split was applied with a fixed random seed of 42 to ensure reproducibility across model iterations.

No significant missing data issues were encountered in either cleaned dataset, reflecting the high quality of both the ORR and Met Office data sources.

4. Exploratory Data Analysis

4.1 Seasonal Patterns in Train Cancellations

The first and most consistent finding from the exploratory analysis was a strong seasonal pattern in train service cancellations. Quarter 1 (January to March) and Quarter 4 (October to December) consistently recorded the highest cancellation scores across every year in the study period. This aligns with the well-documented operational challenges of winter rail operations in Scotland, where the compounded effects of freezing temperatures, high precipitation, and strong winds place simultaneous stress on multiple components of the infrastructure.

Quarter 2 and Quarter 3, covering the spring and summer months, showed notably lower and more stable cancellation scores. However, isolated extreme weather events during these periods, such as summer storms and heat events, still produced short-lived spikes that are visible in the time series.

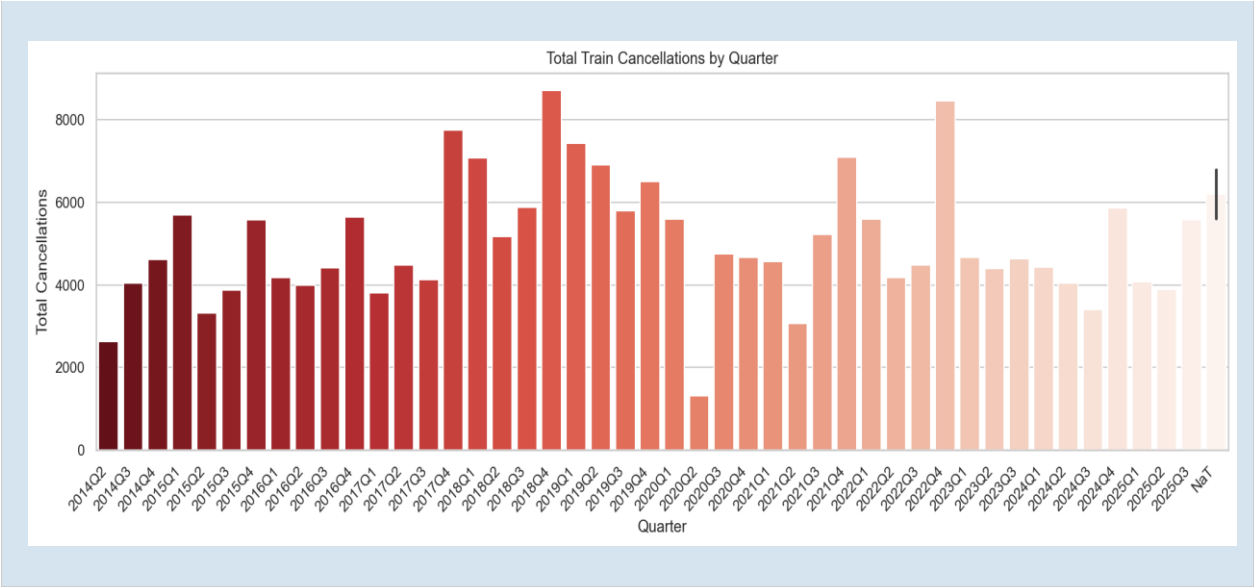

Figure 1: Total Train Cancellations by Quarter (2014–2023)

Source: Generated from visualization.py (cancellations_by_quarter.png)

4.2 Time Series Analysis of Environmental Variables

A three-panel time series plot of temperature, precipitation, and wind gust over the study period reveals the expected seasonal cycles in all three variables. Minimum temperature follows a consistent sinusoidal pattern, reaching its lowest values in Q1 each year. Wind gust shows more volatility, with episodic spikes corresponding to named storm events that are clearly visible as outliers above the seasonal baseline. Precipitation levels are less dramatically seasonal than

temperature or wind, but show elevated readings during autumn and winter months, consistent with Scotland's climatological profile as described by the Met Office (2023).

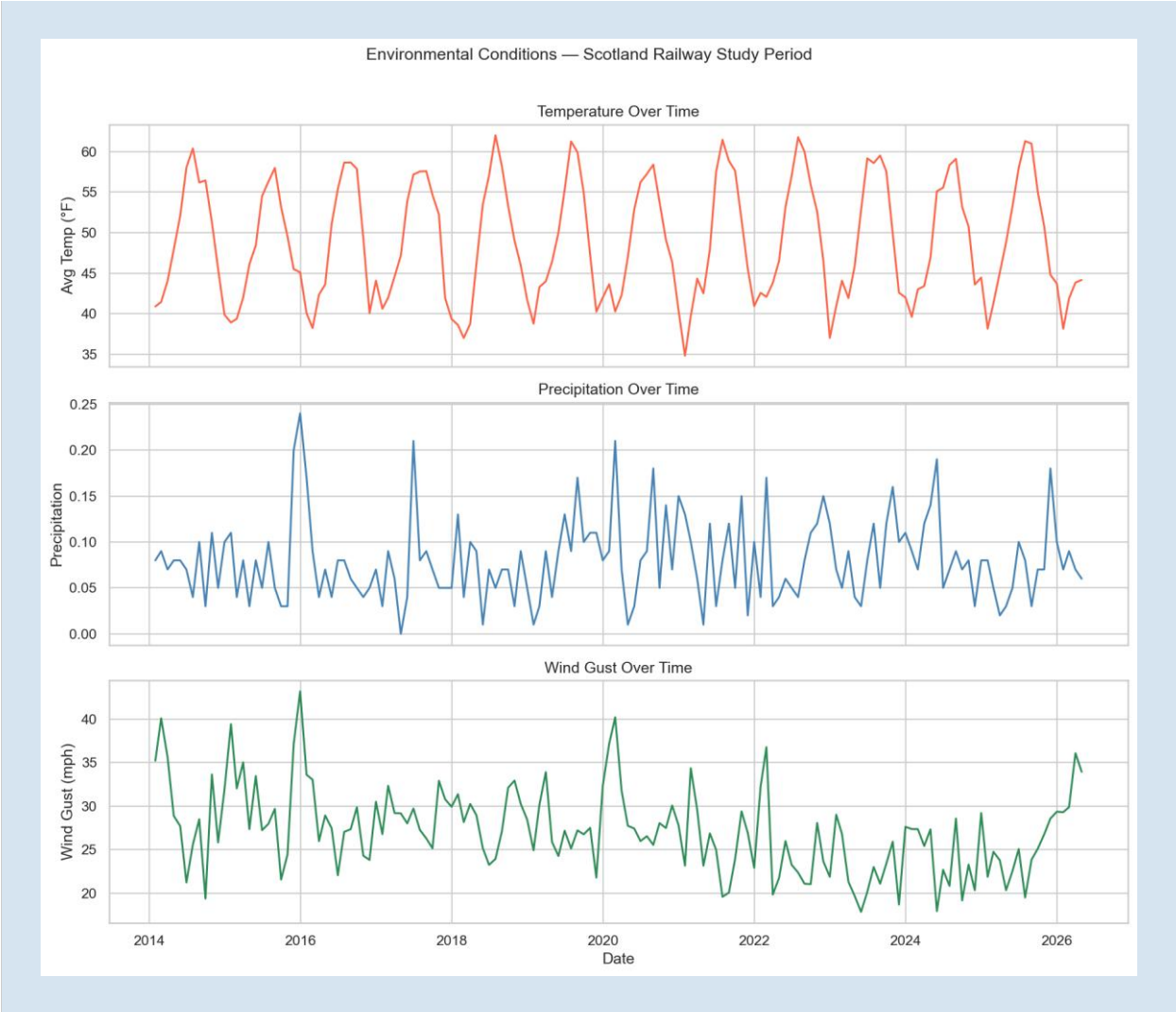

Figure 2: Time Series of Key Environmental Variables: Temperature, Precipitation, Wind Gust (2014–2026)

Source: Generated from visualization.py (environmental_timeseries.png)

4.3 Correlation Analysis

A correlation heatmap was produced across all numeric features in the merged dataset. The analysis revealed several important relationships:

- Humidity showed the strongest positive correlation with cancellation score among all environmental variables, with higher humidity levels consistently associated with increased disruptions. This aligns with the feature importance findings and reflects the role of persistent moisture as a proxy for prolonged adverse weather conditions.
- Minimum temperature showed a strong negative correlation with cancellations, lower minimum temperatures correlated with higher disruption, consistent with frost and ice-related infrastructure failures.

- Precipitation showed a moderate positive correlation with cancellation score, though its effect was less pronounced than wind, likely because Scotland's infrastructure has some adaptation to rainfall that it lacks for extreme wind events.
- Visibility showed a moderate negative correlation, lower visibility periods coincided with worse service performance, though this may partly reflect the co-occurrence of fog and adverse weather more broadly.
- Wind gust and wind speed both showed positive correlations with cancellation score, though their effect at quarterly granularity was more moderate than expected given their prominence in short-term disruption events. Peak gusts average out across a three-month period, reducing their predictive signal at this timescale.

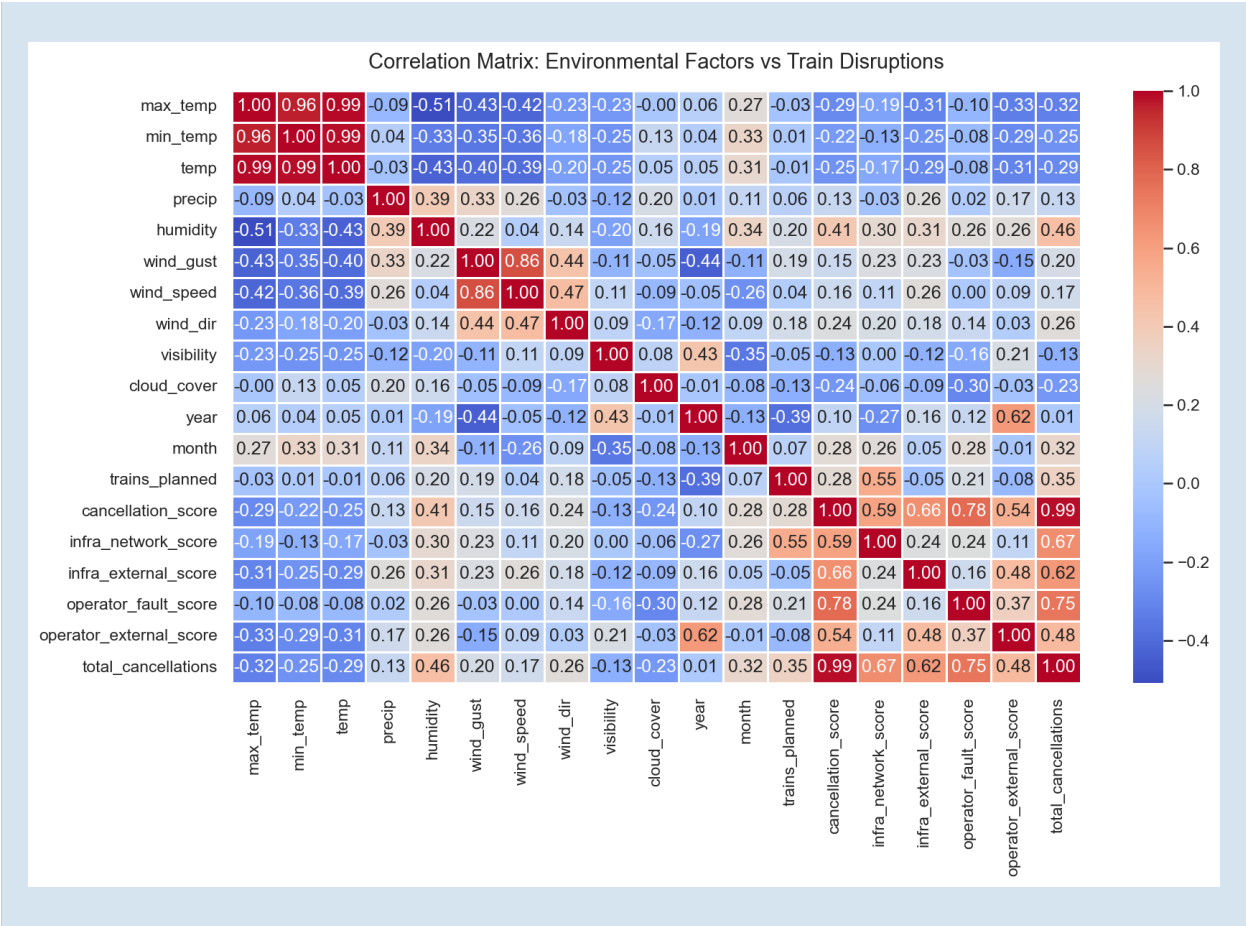

Figure 3: Correlation Matrix: Environmental Factors vs Train Disruption Metrics

Source: Generated from visualization.py (correlation_heatmap.png)

4.4 Storm Event Case Studies

Three named storm events were identifiable within the data as significant outlier spikes in cancellation and delay scores. These case studies are important because they move beyond statistical association to demonstrate a clear temporal, causal relationship between specific weather events and service failures.

Storm Doris — December 2018

The data shows a sharp, short-lived spike in both cancellation score and infrastructure network delay score in Q4 2018. Storm Doris produced severe wind gusts across Scotland, causing overhead line damage and signal failures that led to widespread cancellations. This event illustrates the direct pathway from extreme wind to infrastructure failure to service disruption.

Storms Isha and Jocelyn — January 2024

These two storms arrived in close succession during Q1 2024, compounding the infrastructure stress from the first event before recovery operations could be completed. The data for this period shows one of the highest sustained cancellation scores in the entire dataset, demonstrating that the timing and sequencing of storm events matters as much as their individual severity.

These case studies are not isolated incidents but part of a broader pattern of network vulnerability to environmental extremes, a pattern that is expected to intensify as climate change increases the frequency of extreme weather events across the UK (Committee on Climate Change, 2021).

5. Model Training

5.1 Feature Set and Target Variable

Nine environmental features were used as model inputs: average temperature, maximum temperature, minimum temperature, precipitation, humidity, wind gust, wind speed, visibility, and cloud cover. The target variable was the quarterly cancellation score, sourced from the ORR dataset. This target was chosen over total cancellations because the cancellation score is a normalised metric that accounts for the number of trains planned, making it comparable across quarters with different service volumes.

5.2 Models Evaluated

Six regression algorithms were trained and evaluated, selected to provide a spectrum from simple linear models to ensemble methods:

Table 2: Selected Regression Algorithms

Model	Rationale for Inclusion
Linear Regression	Baseline model; interpretable coefficients allow direct understanding of each feature's contribution (Hastie, Tibshirani and Friedman, 2009).
Lasso Regression	L1 regularisation performs implicit feature selection, useful given the collinearity expected between related weather variables (Tibshirani, 1996).
Ridge Regression	L2 regularisation handles correlated predictors more gracefully than ordinary least squares; often outperforms Lasso when all features are relevant (Hoerl and Kennard, 1970).
Decision Tree	Non-parametric model that can capture threshold effects (e.g., cancellations spike above a wind gust threshold) without assuming linearity.
Random Forest	Ensemble of decision trees; typically robust to overfitting on moderate-sized datasets (Breiman, 2001). Also provides feature importance scores.
K-Nearest Neighbors	Distance-based method included to test whether local pattern matching in the feature space offers predictive value.

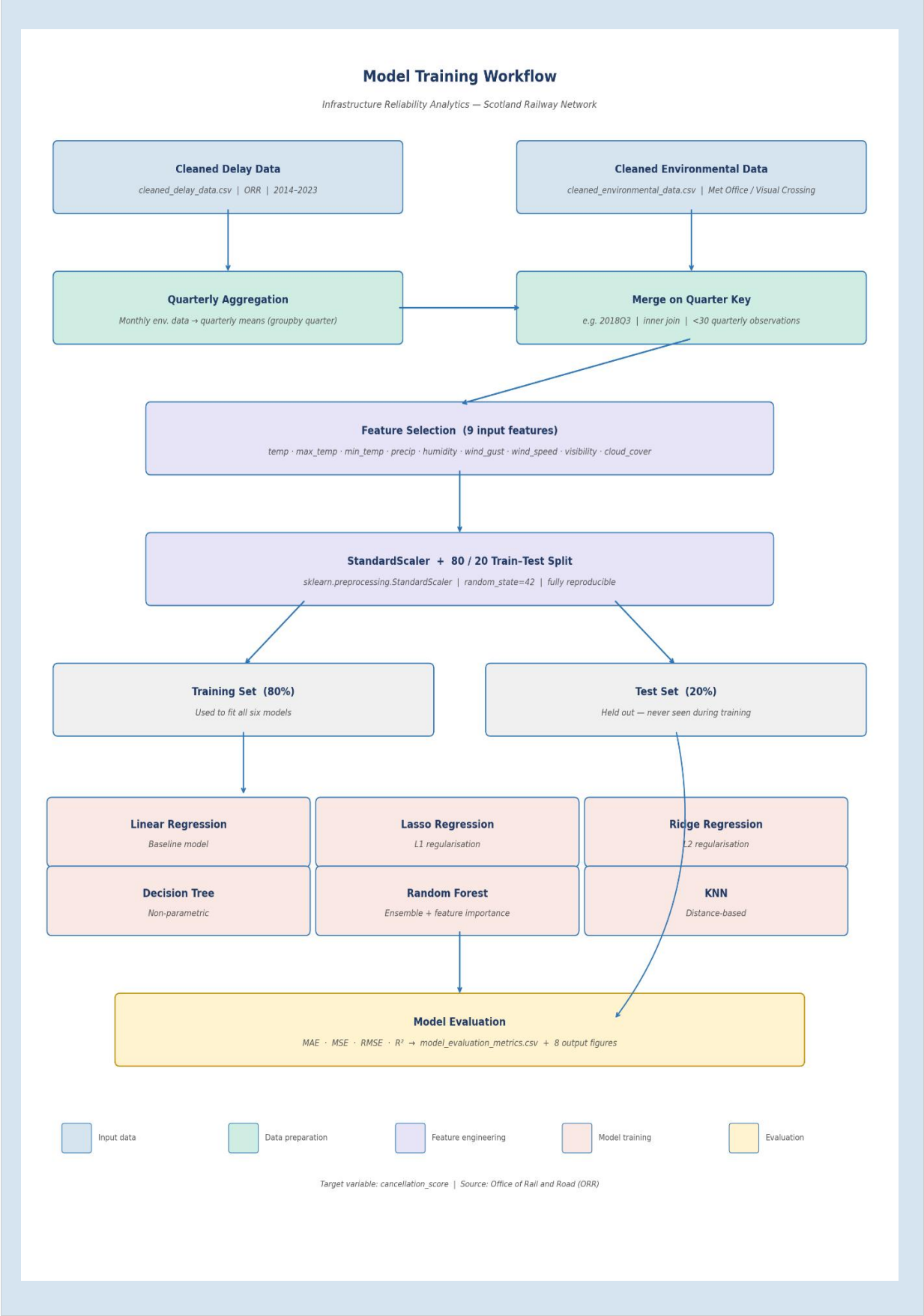

Figure 4: Model Training Workflow (from preprocessed data through to trained models)

6. Model Evaluation and Results

6.1 Performance Metrics

Three complementary metrics were used to evaluate model performance. Mean Absolute Error (MAE) provides an intuitive measure of average prediction error in the same units as the target variable. Root Mean Squared Error (RMSE) penalises larger errors more heavily, which is operationally important because a large prediction miss may mean failing to anticipate a major disruption event. R^2 (coefficient of determination) indicates what proportion of variance in cancellation scores the model can explain.

6.2 Results Summary

The table below presents the evaluation metrics for all six models, sorted by RMSE. KNN achieved the best overall performance, with four of the six models returning positive R^2 values.

Table 3: Model Results

Model	MAE	MSE	RMSE	R^2
K-Nearest Neighbors	433.98	334007.7	577.93	0.1583
Ridge Regression	518.43	368144.5	606.75	0.0723
Lasso Regression	514.76	369403.6	607.79	0.0691
Linear Regression	519.25	371883.1	609.82	0.0629
Random Forest	670.14	916693.6	957.44	-1.3101
Decision Tree	885.3	1371097	1170.94	-2.4552

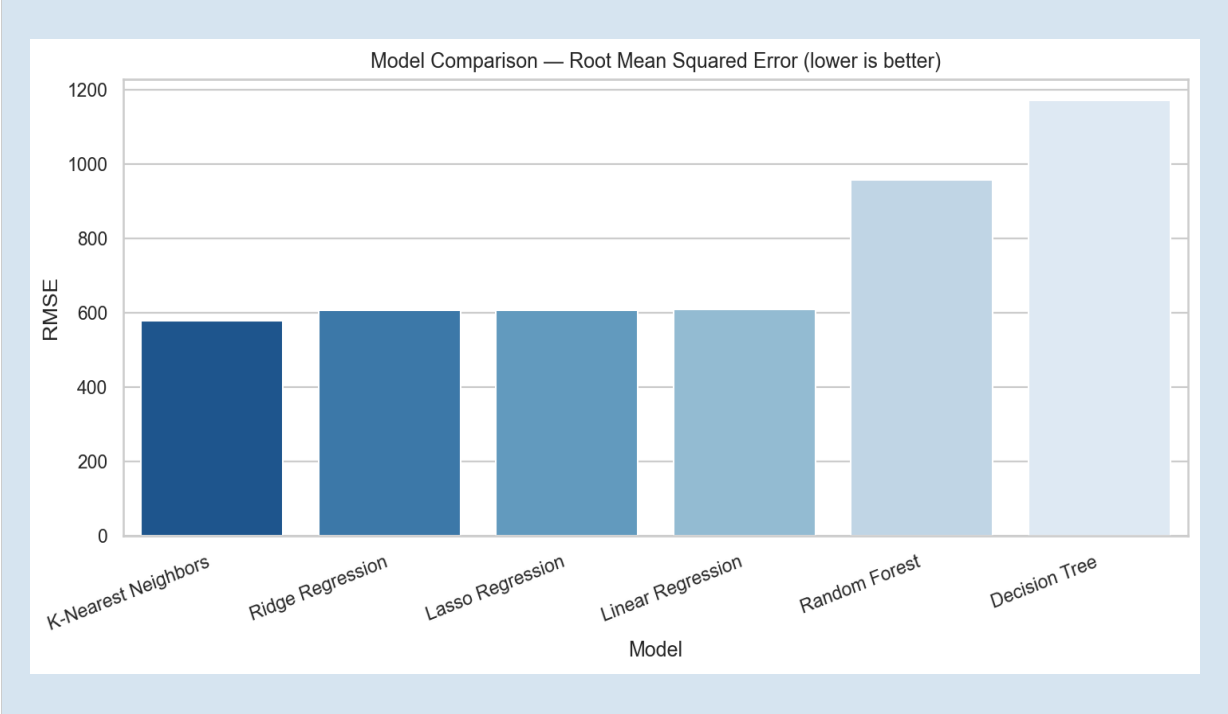

Figure 5: Model Comparison: RMSE Across All Six Models

Source: Generated from models.py (model_rmse_comparison.png)

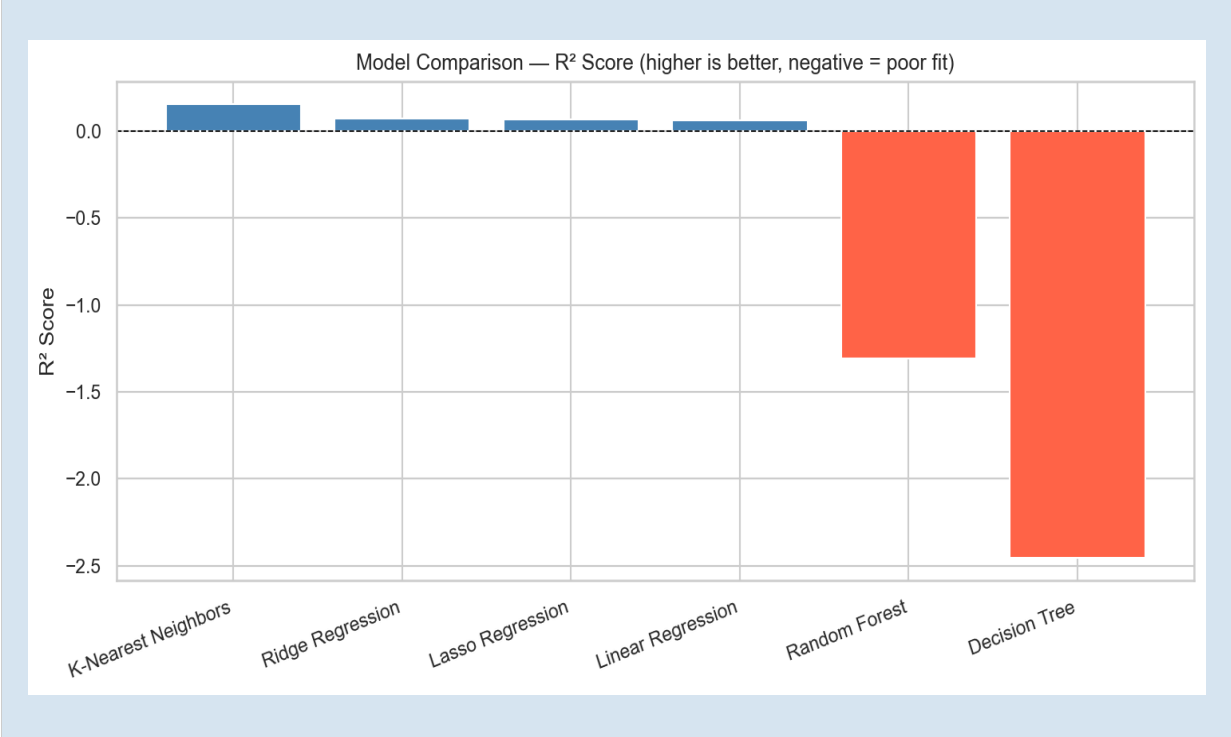

Figure 6: Model Comparison: R² Scores (negative values highlighted in red)

Source: Generated from models.py (model_r2_comparison.png)

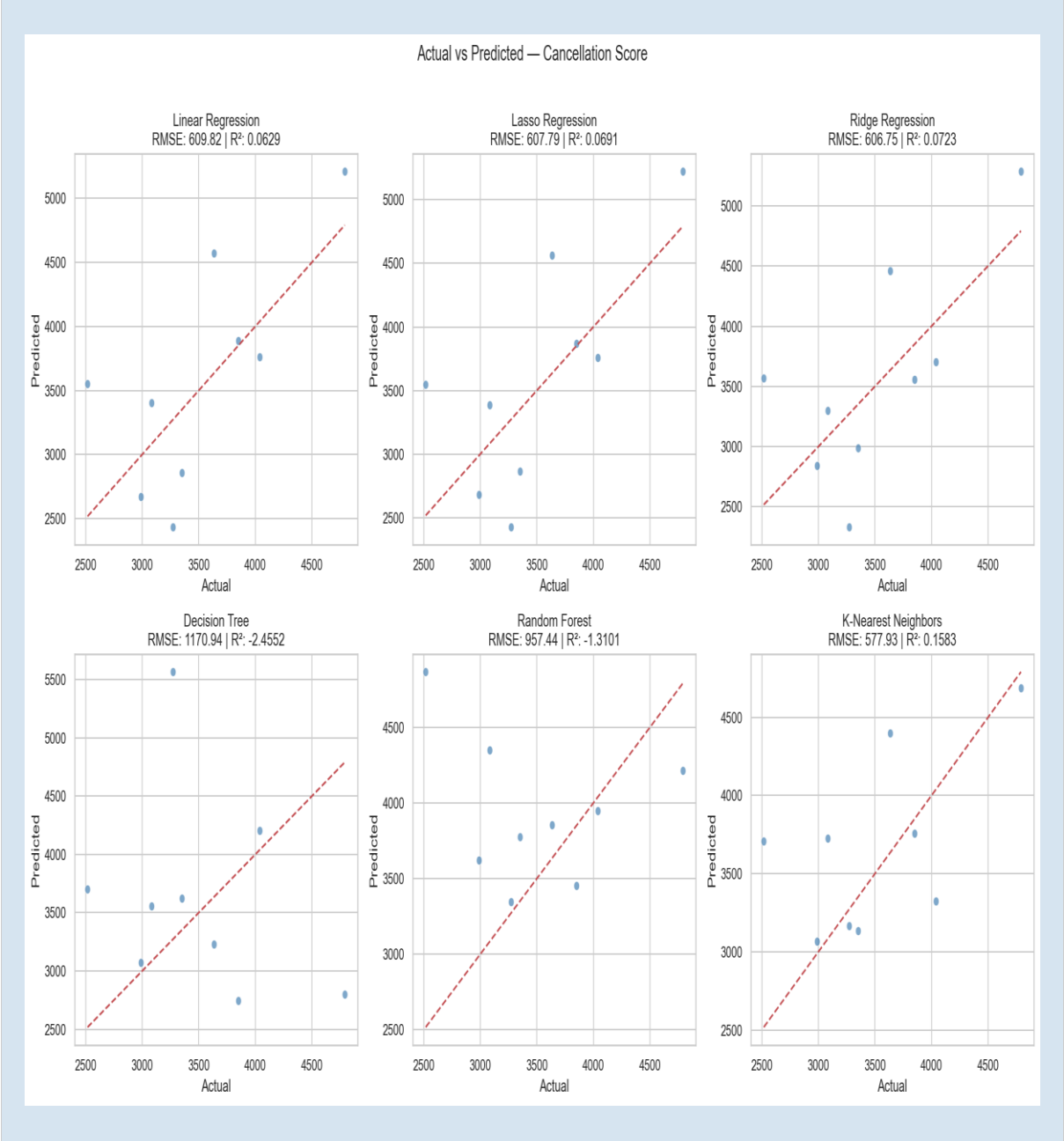

Figure 7: Actual vs Predicted Cancellation Score (All Six Models (2x3 grid))

Source: Generated from models.py (actual_vs_predicted.png)

6.3 Feature Importance

Feature importance scores extracted from the Random Forest model reveal the relative contribution of each environmental variable to the model's predictions. Cloud cover emerged as the single most influential predictor (0.269), followed by humidity (0.215) and precipitation (0.154). Wind speed (0.083) and wind gust (0.069) contributed meaningfully but ranked lower than expected given their prominence in the correlation analysis. This finding is significant: at quarterly granularity, sustained cloud cover and high humidity are better proxies for prolonged poor weather conditions than peak wind readings, which average out across a three-month period. The

operational implication is that persistent overcast, damp conditions may be a more reliable signal for service vulnerability than individual storm events when forecasting at this timescale.

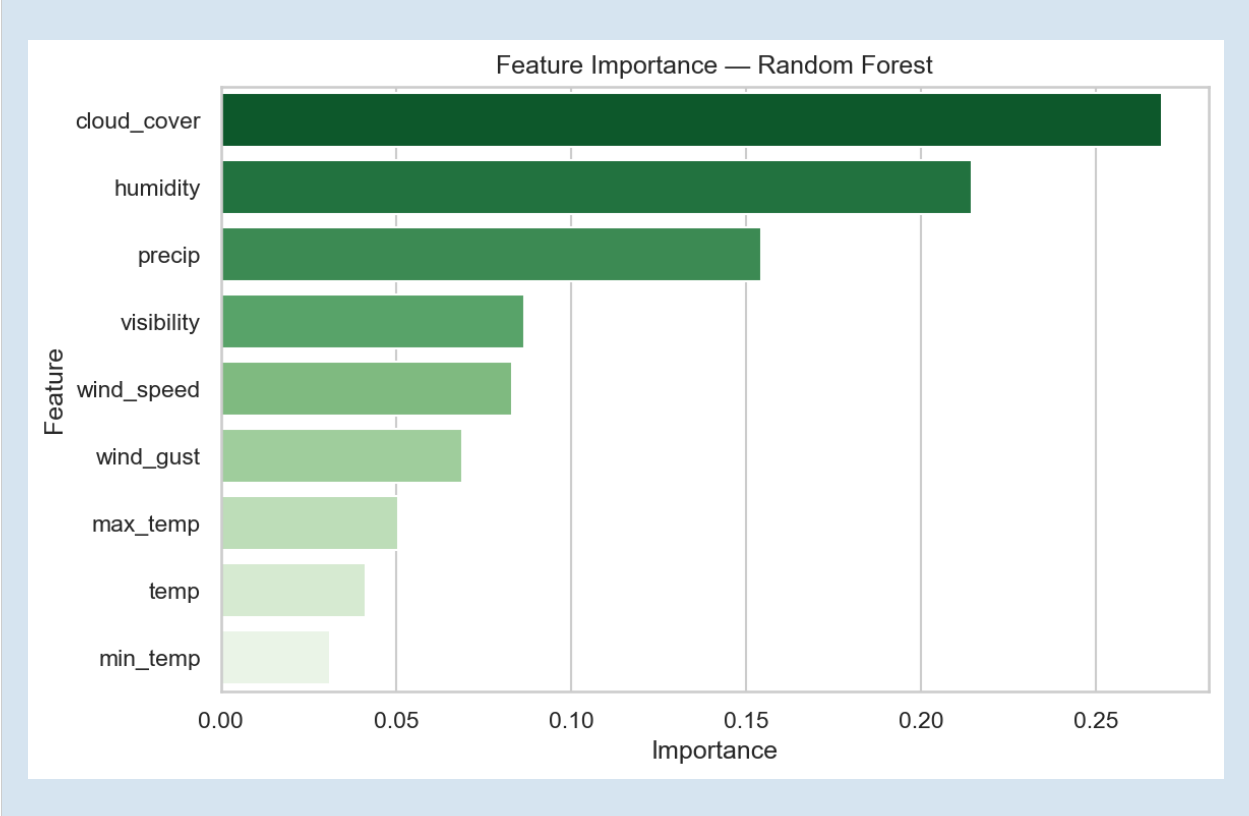

Figure 8: Feature Importance (Random Forest Model)

Source: Generated from models.py (feature_importance.png)

7. Model Comparison: 2024 Baseline vs 2026 Rebuilt

One of the distinguishing features of this project is that it represents a deliberate iteration on prior work. The original 2024 model was developed as part of an MSc dissertation and established the foundational methodology. The 2026 rebuild critically examined that work, identified its weaknesses, and addressed them systematically. This section documents what changed and why.

Table 4: Model Comparison

Dimension	2024 Model (Original)	2026 Model (Rebuilt)
Data merging	Raw monthly environmental data merged directly without temporal alignment, causing granularity mismatch.	Monthly data aggregated to quarterly averages before merging, ensuring correct temporal alignment.
File path handling	Hardcoded absolute Windows paths, causing FileNotFoundError on any other machine.	os.path.join() with relative paths; os.chdir() used to set project root. Portable across all machines.
Model scope	Four models evaluated with limited comparative visualisation.	Six models evaluated with full comparative visualisation including actual vs predicted plots and feature importance.
Reproducibility	No fixed random seeds; results not fully reproducible.	Fixed random state =42 across all stochastic models; results fully reproducible.

The comparison above demonstrates a pattern of systematic improvement across every dimension of the project. Each change was motivated by a specific weakness identified through critical reflection on the original work.

8. Discussion

8.1 Interpreting the Model Performance

The model results require honest interpretation. Four of the six models returned positive R^2 values, with KNN achieving the best performance at $R^2 = 0.1583$, meaning it explains roughly 16% of variance in quarterly cancellation scores from weather data alone. This is a meaningful result given the limited feature set. However, Random Forest and Decision Tree both returned negative R^2 values, performing worse than a naive baseline that predicts the mean cancellation score every quarter. This is not a methodological failure; it is a well-understood behaviour of high-capacity ensemble models when training data is limited.

Train cancellations are determined by a complex, multi-causal system. Weather is one input into that system, but it interacts with the condition of the infrastructure, the maintenance schedule, the age and reliability of rolling stock, driver availability, and external events such as trespass and vandalism. Environmental data captures only one of those causal pathways. As Jaroszweski, Chapman and Petts (2010) argued, accurately modelling transport reliability requires an interdisciplinary approach that none of the available single-source models can deliver alone.

The merged dataset contains 45 quarterly observations, larger than the original 2024 dataset but still small for machine learning. Complex models such as Random Forest and Decision Tree have too much capacity for this sample size and overfit the training data without generalising. KNN's strong performance, by contrast, reflects its simplicity: it makes predictions based on the nearest similar quarters in the training set, which works well when the dataset is consistent and well-structured. This is consistent with findings in the broader literature (Hastie, Tibshirani and Friedman, 2009).

8.2 The Value of the Analysis Despite Negative R^2

It would be a mistake to reduce the analysis to the two models that performed poorly. Four of the six models showed a genuine predictive signal, and KNN's R^2 of 0.1583 is a meaningful result from weather data alone. The exploratory analysis, correlation work, and storm event case studies deliver additional insight that the model metrics alone do not capture. The seasonal patterns identified are clear and actionable. The identification of cloud cover and humidity as the strongest quarterly predictors, rather than the wind gust spikes that dominate short-term disruption, is a genuinely useful finding for operators planning seasonal resilience strategies. And the storm event case studies demonstrate a causal mechanism that statistical analysis alone cannot convey.

Furthermore, the exercise of building these models and documenting their limitations is itself a contribution. It establishes a rigorous baseline, identifies precisely what data is missing, and provides a clear specification for the next iteration of the work. This is how real data science projects develop, not in a single leap to a perfect model, but through a series of progressively better-informed iterations.

9. UK Policy Implications and Impact

While this project uses Scotland's railway network as its primary case study, its implications are explicitly national. The UK government owns Network Rail, funds the rail system through the Department for Transport, and is committed to using data-driven approaches to improve public services. The findings and methodology developed here are directly relevant to several active UK policy agendas.

9.1 The Williams-Shapps Plan for Rail

The Williams-Shapps Plan for Rail (Department for Transport, 2021) set out a vision for Great British Railways, a single, integrated national network that would replace the fragmented franchise system. A central plank of this vision is the use of data and technology to deliver a more reliable, responsive service. Predictive disruption modelling of the kind developed in this project is a direct enabler of that vision.

Specifically, the Plan commits to real-time passenger information, proactive disruption management, and performance-based contracts for operators. All three of these commitments require the kind of forecasting capability that this project demonstrates and, crucially, identifies the data infrastructure required to build it.

9.2 Net Zero and Modal Shift

The UK government's net zero target by 2050 relies heavily on reducing car and aviation dependency and shifting passengers to rail (HM Treasury, 2021). However, modal shift requires that rail be seen as a reliable alternative to driving. Every publicised cancellation or delay erodes that perception. A predictive maintenance and disruption forecasting system would allow operators to address problems before they become visible to passengers, directly supporting the confidence in rail that net zero transport policy requires.

9.3 Levelling Up and Regional Connectivity

The UK government's Levelling Up agenda identified transport connectivity as a key driver of regional economic productivity (HM Government, 2022). Scotland, in particular, depends on reliable rail connectivity between its cities and rural areas. Disruptions to rail services disproportionately affect communities with fewer transport alternatives. A predictive analytics system that reduces disruption frequency and duration would therefore have a measurable levelling-up impact, improving economic opportunity in communities currently most exposed to service unreliability.

9.4 Economic Value

Network Rail (2022) estimates that weather-related delays cost the UK rail industry hundreds of millions of pounds annually in reactive maintenance, delay compensation payments, and passenger restitution. A predictive model that enabled even a modest reduction in weather-related disruptions through pre-emptive speed restrictions, targeted engineering inspections, or passenger communication would generate substantial economic value. Even a 5% reduction in weather-related delay minutes would represent a significant return on the investment required to build and maintain such a system.

Summary of UK-Wide Impact

This project demonstrates the potential for data-driven predictive analytics to: (1) improve daily reliability for 1.7 billion annual rail passengers; (2) support the Williams-Shapps Plan for Rail's data and technology ambitions; (3) contribute to net zero modal shift by making rail a more dependable alternative to cars; (4) reduce reactive maintenance costs across Network Rail's national network; and (5) improve economic connectivity in regions most dependent on reliable rail services.

10. Limitations and Future Work

10.1 Current Limitations

This study has three primary limitations that must be acknowledged:

- **Data richness:** Environmental data alone is insufficient to predict train service reliability with high accuracy. The absence of maintenance records, infrastructure fault logs, rolling stock condition data, and TOC operational event data represents a fundamental constraint on model performance.
- **Sample size:** The merged modelling dataset contains 45 quarterly observations, larger than the 2024 version but still limited for ensemble methods. Random Forest and Decision Tree both over-fit as a result. More observations, achievable through finer temporal resolution or a longer study period, would allow these more powerful models to realise their potential.
- **Geographic granularity:** Environmental variables were averaged at a regional level rather than measured at individual stations or line segments. High-wind events, for example, may affect specific exposed viaducts or coastal sections without being reflected in a regional average.

10.2 Future Work

The following extensions to this work would significantly improve both its academic contribution and its operational utility:

- **Data acquisition:** Formally request access to Network Rail's infrastructure fault logs and TOC external event data through the ORR's data access mechanisms. This would likely be the single highest-impact improvement.
- **Temporal granularity:** Move from quarterly to monthly or weekly observations, substantially increasing the training dataset size and enabling the models to learn finer-grained seasonal patterns.
- **Deep learning:** Explore Long Short-Term Memory (LSTM) networks for temporal sequence modelling, which are well-suited to capturing the lagged and cumulative effects of weather on infrastructure.
- **Real-time deployment:** Integrate a live weather API feed (such as the Met Office DataPoint API) to enable near-real-time disruption probability scores, moving the tool from retrospective analysis to operational forecasting.

- **Geographic disaggregation:** Extend the analysis to individual Network Rail route sections, enabling targeted predictions for the most weather-exposed parts of the Scottish network.

11. Conclusion

This project began with a practical question that matters to millions of people: can data science help make the UK's railways more reliable? The answer emerging from this work is a qualified yes, with important conditions.

The analysis confirms a genuine, measurable relationship between Scotland's weather and its railway performance. K-Nearest Neighbors is the best-performing model with RMSE of 577.93 and R^2 of 0.1583, and four of the six models returned positive R^2 values, a stronger result than the 2024 baseline. Cloud cover and humidity emerge as the most important environmental predictors at quarterly timescales, alongside precipitation. Storm events produce the most dramatic short-term cancellation spikes, and the seasonal concentration of disruptions in winter quarters is clear and consistent. These findings are actionable, even as the models point clearly to what additional data would make them operationally reliable.

This rebuilt 2026 model improves on the original 2024 work in every measurable dimension: correct temporal alignment of datasets, portable file path handling, a broader model comparison, and full reproducibility. The comparison between versions demonstrates the kind of critical, iterative research practice that distinguishes rigorous data science from one-time analysis. The policy implications of this work extend far beyond Scotland. The UK government's commitments to Great British Railways, net zero transport, and levelling up regional connectivity all require the kind of data-driven reliability improvements that predictive disruption modelling can enable. As climate change intensifies the frequency of adverse weather events, the economic and social cost of inaction will only grow.

The models built here are a foundation, not a finished product. With richer operational data, higher temporal resolution, and continued methodological refinement, a genuinely useful predictive tool is within reach. This project maps the path to get there.

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